

A seminorm increment bound passes to every derivative direction

MathlibAnnex.FDeriv.norm_apply_le_seminorm_of_lipschitz

theorem

Takes a difference-quotient limit without imposing a positive lower bound.

Statement

Let $n, N \in \mathbb{N}_0$, let $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ have the sup norms, and let $p : E \rightarrow \mathbb{R}$ be a real seminorm. Let $f : E \rightarrow F$ satisfy

$$\|f(y) - f(z)\|_\infty \leq p(y - z) \quad (y, z \in E).$$

Fix a point $x \in E$ at which f is Fréchet differentiable, and write $Df(x)$ for its derivative. Then

$$\|Df(x)v\|_\infty \leq p(v) \quad (v \in E).$$

Assumptions

The seminorm may be degenerate. Neither norm equivalence nor a positive lower comparison is assumed. The increment estimate holds globally; differentiability is assumed only at the chosen point. Both dimensions may be zero.

Conclusion

The same seminorm bounds $Df(x)v$ for every direction, including zero.

Let λ_n denote coordinate Lebesgue measure on E . The separate almost-everywhere wrapper additionally takes $U \geq 0$ with $p(z) \leq U\|z\|_\infty$. Then $\|f(y) - f(z)\|_\infty \leq U\|y - z\|_\infty$, so Rademacher gives differentiability almost everywhere for reference Lebesgue measure. Restriction to any set $s \subseteq E$ and the pointwise result give the bound for every v , for $\lambda_n|_s$ -almost every x . The extra upper comparison is an input of this wrapper.

Proof route

Apply the increment estimate on one affine line, then pass to its derivative.

▼ Proof steps

1. **Estimate one directional quotient.** Fix $v \in E$ and let $\phi(t) = f(x + tv)$. The increment assumption and seminorm homogeneity give

$$\|\phi(t) - \phi(0)\|_\infty \leq p(tv) = |t|p(v).$$

For every real $t \neq 0$, division by $|t| > 0$ gives

$$\left\| \frac{f(x + tv) - f(x)}{t} \right\|_{\infty} \leq p(v).$$

This also applies to $v = 0$.

2. Take the derivative limit. Differentiability at x gives

$$\frac{f(x + tv) - f(x)}{t} \longrightarrow Df(x)v \quad (t \rightarrow 0, t \neq 0).$$

Continuity of the norm therefore yields

$$\|Df(x)v\|_{\infty} = \lim_{t \rightarrow 0, t \neq 0} \left\| \frac{f(x + tv) - f(x)}{t} \right\|_{\infty} \leq p(v).$$

The source's local derivative-bound argument uses ϕ , $d = Df(x)v$ and $C = p(v)$. The previous estimate and this derivative limit are its inputs; its output is exactly the asserted inequality.

Main citations

- Exact declaration and its source — `MathlibAnnex.FDeriv.norm_apply_le_seminorm_of_lipschitz`
- The separate almost-everywhere wrapper —
`MathlibAnnex.FDeriv.ae_norm_apply_le_seminorm_of_lipschitz`

Lean source signature (exact)

```
/-- A seminorm increment bound passes to the derivative in each direction. -/
theorem norm_apply_le_seminorm_of_lipschitz {n N : ℕ}
  (p : Seminorm ℝ (Fin n → ℝ)) {f : (Fin n → ℝ) → (Fin N → ℝ)}
  (hf : ∀ x y, ‖f x - f y‖ ≤ p (x - y)) {x : Fin n → ℝ}
  (hx : DifferentiableAt ℝ f x) : ∀ v, ‖(fderiv ℝ f x) v‖ ≤ p v
```

Read `hf` and `hx` as the two hypotheses; neither is an additional map.

In the source	Mathematical meaning
<code>{n N : ℕ}</code>	Arbitrary nonnegative integers n, N . Braces mean that Lean may infer these parameters; they are not extra assumptions.
<code>Fin n → ℝ; Fin N → ℝ</code>	The spaces $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ with their sup norms. A vector is a list of real coordinates, indexed from 0 in Lean and from 1 in the formulas here.
<code>p : Seminorm ℝ (Fin n → ℝ)</code>	A real seminorm $p : E \rightarrow \mathbb{R} : p(x) \geq 0$, $p(x + y) \leq p(x) + p(y)$ and $p(tx) = t p(x)$. It may vanish at a nonzero vector.

In the source	Mathematical meaning
<code>{f : ... → ...}; {x : Fin n → ℝ}</code>	The function $f : E \rightarrow F$ and the chosen point $x \in E$. Braces let Lean infer these inputs.
<code>hf : ∀ x y, ‖f x - f y‖ ≤ p (x - y)</code>	The global hypothesis $\ f(y) - f(z)\ _\infty \leq p(y - z)$ for every $y, z \in E$. The bound variables called x, y here range independently of the later chosen point x .
<code>hx : DifferentiableAt ℝ f x</code>	The function f is Fréchet differentiable over $\mathbb{R}$ at the chosen point x .

In the source	Mathematical meaning
<code>(fderiv ℝ f x) v</code>	Apply the linear derivative $Df(x)$ to the direction $v \in E$; this is the vector $Df(x)v$, not a scalar derivative.
<code>∀ v, ‖(fderiv ℝ f x) v‖ ≤ p v</code>	The conclusion $\ Df(x)v\ _\infty \leq p(v)$ for every $v \in E$. There is no almost-everywhere qualifier or measure in this theorem.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 17–22:

```
noncomputable section

open Set MeasureTheory Filter
open scoped Topology

namespace MathlibAnnex.FDeriv
```

▼ Exact content identity

Declaration: `MathlibAnnex.FDeriv.norm_apply_le_seminorm_of_lipschitz`

Accepted content SHA-256: `5b41411f5dc056e27160fcebab5aa5e88e29a47ac8906cf19dabab8f075a2596`

Accepted source guide SHA-256: `0ee2158dfd51abb0dd4d0afc25dfcfcfada4614c2964db213f54bb54f75f398`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`